Detect, Investigate, Judge and Determine: A Novel LLM-based Framework for Few-shot Fake News Detection

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Abstract

Few-Shot Fake News Detection (FS-FND) aims to distinguish inaccurate news from real ones in extremely low-resource scenarios. This task has garnered increased attention due to the widespread dissemination and harmful impact of fake news on social media. Large Language Models (LLMs) have demonstrated competitive performance with the help of their rich prior knowledge and excellent in-context learning abilities. However, existing methods face significant limitations, such as the Understanding Ambiguity and Information Scarcity, which significantly undermine the potential of LLMs. To address these shortcomings, we propose a Dual-perspective Augmented Fake News Detection (DAFND) model, designed to enhance LLMs from both inside and outside perspectives. Specifically, DAFND first identifies the keywords of each news article through a Detection Module. Subsequently, DAFND creatively designs an Investigation Module to retrieve inside and outside valuable information concerning to the current news, followed by another Judge Module to derive its respective two prediction results. Finally, a Determination Module further integrates these two predictions and derives the final result. Extensive experiments on two publicly available datasets show the efficacy of our proposed method, particularly in low-resource settings.

1 Introduction

Fake News Detection (FND), aiming to distinguish between inaccurate news and legitimate news, has garnered increasing importance and attention due to the the pervasive dissemination and detrimental effects of fake news on social media platforms (Shu et al., 2017). Few-Shot Fake News Detection (FS-FND), as a subtask of FND, endeavors to identify fake news by leveraging only K instances per category (K-shot) in the training phase (Hu et al., 2024; Gao et al., 2021; Ma et al., 2023).



Figure 1: An example of fake news detection and the comparison between existing LLM-based methods and our design.

Generally, fake news detection can be framed as a binary classification problem and addressed using various classification models. In the early stage, researchers primarily employ machine learning or deep learning algorithms to represent and classify candidate news articles (Horne and Adali, 2017; Jiang et al., 2022). More recently, with the advent of Large Language Models (LLMs), FS-FND has been effectively addressed through the In-Context Learning (ICL) technology, which is particularly prevalent in few-shot settings (Hu et al., 2024; Boissonneault and Hensen, 2024). Among them, Hu et al. (2024); Wang et al. (2024); Teo et al. (2024) were pioneers in investigating the potential of LLMs in this field.

However, this kind of methods mainly directly ask the LLM to judge the authenticity of the given news, which often exceeds the capabilities of LLMs in many circumstances, particularly those relatively small LLMs in common usage (e.g., 7B param-

eters). An example of this can be observed in Figure 1, which presents a news about microchip developments. Existing LLM-based approaches encounter two principal challenges: (1) Understanding Ambiguity: LLMs may fail to understand and grasp the core meaning conveyed in the news, thereby straining the detection process. (2) Information Scarcity: Due to the timeliness nature of news content, the training corpus of LLMs is frequently outdated. This poses fundamental difficulties in the detection of fake news.

To this end, this paper proposes a novel approach to address the two aforementioned issues. Specifically, to mitigate the Understanding Ambiguity problem, we aim to extract valuable insights from an inside perspective by retrieving similar samples from the training set, thereby enhancing the comprehension of key concepts in the target news. Concurrently, to tackle the Information Scarcity problem, we employ an external search engine to gather relevant information about the news online. This integrates real-time data, effectively overcoming the limitation of information obsolescence.

More specifically, we design a Dual-perspective Augmented Fake News Detection (DAFND) model. DAFND comprises four key components: (a) A Detection Module: This module employs LLMs to extract keywords from each news article, providing an effective query for the subsequent modules. (b) An Investigation Module: It investigates more valuable information related to the target news, which comes from both inside (i.e., training set) and outside (i.e., search engine) perspectives. (c) A Judge Module: This module designs prompts that enable LLMs to generate predictions and explanations based on the findings from both inside and outside investigations. (d) A Determination Module: It takes into account the predictions and explanations from both two perspectives, and thus makes a final decision with high confidence, especially in cases where two perspectives conflict.

In summary, the main contributions of our work could be summarized as follows.

- For the first time, we explore augmenting LLMs from inside and outside perspectives for few-shot fake news detection, pioneering a novel direction in this field.
- We devise the Dual-perspective Augmented Fake News Detection (DAFND) model, which effectively addresses the Understanding Ambiguity and Information Scarcity problems, particularly in low-resource settings.
- We conduct extensive experiments on two publicly available datasets, where the experimental results demonstrate the effectiveness of our proposed method. We will make our codes publicly available upon acceptance of the paper.

2 Related Work

2.1 Few-Shot Fake News Detection

The objective of fake news detection task is to distinguish inaccurate news from real ones (Shu et al., 2017). For few-shot fake news detection, only K instances per category (K-shot) are sampled for the training phase (Gao et al., 2021; Ma et al., 2023).

Generally, fake news detection can be defined as a binary classification problem and is addressed by a variety of classification models. Initially, researchers mainly rely on feature engineering and machine learning algorithms. For example, Horne and Adali (2017) presented a set of content-based features to a Support Vector Machine (SVM) classifier. With the rapid development of computing power, significant improvements have been made with the help of various deep learning algorithms and Pre-trained Language Models (PLMs). Ghanem et al. (2021) combined lexical features and a Bi-GRU network to achieve accurate fake news detection. Jiang et al. (2022) proposed Knowledgeable Prompt Learning (KPL), incorporating prompt learning into fake news detection for the first time, and achieved state-of-the-art performance.

Additionally, due to the specificity of news articles, researchers have incorporated external knowledge knowledge to contribute to traditional fake news detection models. For instance, Dun et al. (2021); Ma et al. (2023); Hu et al. (2021b) utilized knowledge graphs to enrich entity information and structured relation knowledge, leading to more precise news representations and improved detection performance. Meanwhile, Huang et al. (2023) adopted a data augmentation perspective, proposing a novel framework for generating more valuable training examples, which has proven to be beneficial in detecting human-written fake news.

Currently, with the advent of large language models, many researchers are exploring few-shot fake news detection through in-context learning and data augmentation technologies (Hu et al., 2024; Wang et al., 2024; Teo et al., 2024). However, These methodologies either simply apply LLMs to judge the authenticity of the given news, or employ

LLMs to rephrase the training data, thereby not fully utilizing the potential of LLMs. More importantly, most of them are significantly limited by the aforementioned two shortcomings, particularly in the Information Scarcity problem.

2.2 Large Language Models

The emergence of Large Language Models (LLMs) such as GPT-4, LLama-3 and others (Hoffmann et al., 2022; OpenAI, 2023; AI@Meta, 2024; Tunstall et al., 2023), marks a significant advancement in the field of natural language processing. In-context learning, a novel few-shot learning paradigm, was initially introduced by Brown et al. (2020). To date, LLMs have exhibited remarkable performance across a range of NLP tasks, including text classification, information extraction, question answering and fake news detection (Hu et al., 2024; Wang et al., 2024; Teo et al., 2024; Liu et al., 2022; Zhao et al., 2021).

Previous studies (Hu et al., 2024; Wang et al., 2024; Teo et al., 2024) have endeavored to solve few-shot fake news detection via directly asking LLMs or employing them to rephrase the training data. For example, Hu et al. (2024) explored the potential of LLMs in fake news detection, and further developed an Adaptive Rationale Guidance (ARG) network to synergize traditional methods with large language models. Similarly, Wang et al. (2024) leveraged LLMs to generate justifications towards evidence relevant to given news, which were subsequently fed into a trainable classifier.

3 Problem Statement

Generally, fake news detection can be framed as a binary classification problem, wherein each news article is classified as either real ($y = 0$) or fake ($y = 1$) (Dun et al., 2021). Formally, each piece of news S is composed of a sequence of words, i.e., $S = \{s_1, s_2, ..., s_n\}$, encompassing its title, content text, and relevant tweets. The goal is to learn a detection function $F : F(S) \Longrightarrow y$, where $y \in \{0, 1\}$ denotes the ground-truth label of news.

In the few-shot settings, adhering the strategy employed in (Gao et al., 2021; Ma et al., 2023), we randomly sample K instances per category (K-shot)[1] for the training phase. The entire test set is preserved to ensure the comprehensiveness and effectiveness of evaluation.

[1]This implies that for a K-shot fake news detection setting, the number of training instances is $2K$.

4 The DAFND Model

In this section, we will introduce the technical details of DAFND model. As depicted in Figure 2, DAFND comprises of four distinct components: (a) Detection Module; (b) Investigation Module; (c) Judge Module; (d) Determination Module. These modules are sequentially interconnected to achieve the final accurate detection of fake news.

4.1 Detection Module

In this module, we aim to identify the key information contained in the given news article, which will serve as the query for the subsequent modules. Specifically, we construct prompts to convey the original news article to the LLM. The LLM is then guided to extract N keywords $\{w_1, w_2, ..., w_N\}$, which are expected to address the question: "*when*, *where*, *who*, *what*, *how* and *why* did the given news S happen?":

$$\{w_1, ..., w_N\} = \mathcal{F}(Ins_w, S), \tag{1}$$

where $\mathcal{F}$ represents the LLM, and Ins_w denotes the prompt for the in-context learning. We provide a more detailed description about it in Appendix A.1.

4.2 Investigation Module

In this module, we aim to investigate the relevant information to assist the LLM in conducting accurate inferences. As outlined in Section 1, this process is approached from two perspectives: Inside Investigation and Outside Investigation.

Inside Investigation. To address the Understanding Ambiguity problem introduced in the Introduction, we retrieve effective demonstrations to enhance the LLM's understanding during the in-context learning process (Liu et al., 2022; Rubin et al., 2022).

Specifically, we first concatenate the extracted N keywords of each news, and then utilize the pre-trained language model $\mathcal{M}$ to obtain the representation of these keywords $\{w_1, ..., w_N\}$:

$$\begin{aligned} W &= w_1 \uplus w_2 \uplus ... \uplus w_N, \\ H &= \mathcal{M}(W), \end{aligned} \tag{2}$$

where $\uplus$ represents the concatenation operation. The derived representation H is used to represent each news sample. Along this line, we can further obtain the representation and label pairs (H_i, l_i)

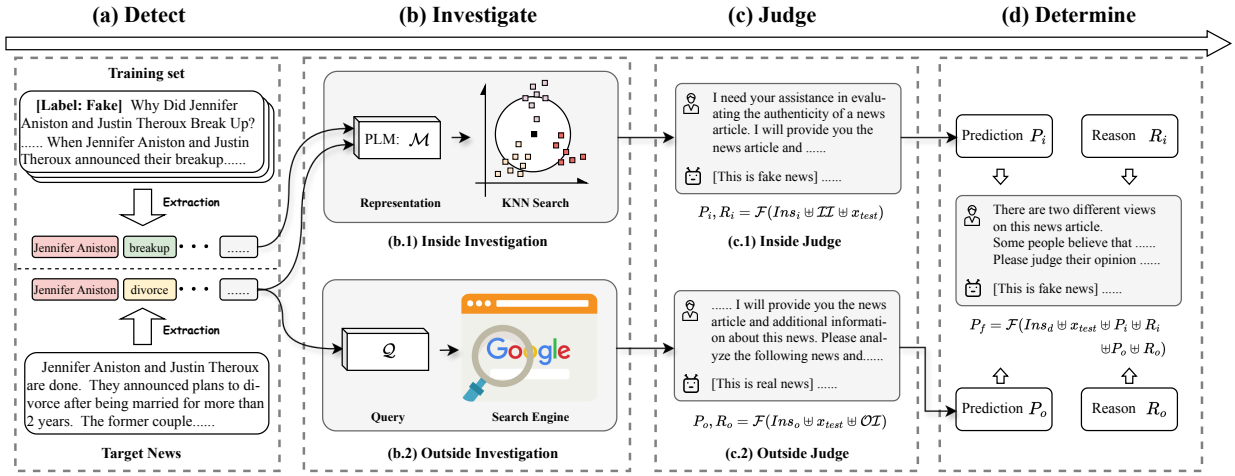


Figure 2: The architecture of our DAFND model. It includes four sequentially connected parts: (a) Detection Module; (b) Investigation Module; (c) Judge Module; (d) Determination Module.

for the training set[2], which constitute a datastore, denoted as D.

Subsequently, when inferring a candidate news j, we employ the k-Nearest Neighbors (kNN) search method (Khandelwal et al., 2019) to retrieve valuable samples from the training set. In detail, we use the representation H_j of news j to query the datastore D according to the euclidean distance. Then, based on the computed distance, we select the nearest k positive and negative news samples, respectively[3]:

$$\mathcal{II} = \{U_{positive}, U_{negative}\}. \quad (3)$$

As a consequence, we obtain the inside investigation outcome $\mathcal{II}$, comprising $2k$ instances.

Outside Investigation. In response to the Information Scarcity problem, we further retrieve additional real-time information from external sources. Inspired by Yoran et al. (2023); Paranjape et al. (2023), we implement a retriever based on the google search engine, using the SerpAPI service[4].

Specifically, based on the extracted keywords $\{w_1, w_2, ..., w_N\}$ in Section 4.1, we first concatenate them to construct the initial query $\mathcal{Q} = w_1 \uplus w_2 \uplus ... \uplus w_N$. Then, following the strategy proposed in (Yoran et al., 2023), we further format the search queries as "`en.wikipedia.org` $\mathcal{Q}$", with the Wikipedia domain preceding the intermediate question. We return the top-1 evidence retrieved by Google. And all retrieved evidence sentences are prepended to the outside investigation outcome, denoted as $\mathcal{OI}$.

4.3 Judge Module

Following the paradigm designed in the Investigation Module (Section 4.2), we attempt to derive the inference results based on the investigated information from both inside and outside perspectives.

Inside Judge. After obtaining the effective demonstrations $\mathcal{II}$ from Inside Investigation, we design prompts to provide the essential information to the LLM, thereby generating the inside prediction. Specifically, inspired by the various attempts about in-context learning (Paranjape et al., 2023), we first describe the target of the fake news detection task through an inside instruction. Then, the retrieved inside investigation results $\mathcal{II} = \{U_{positive}, U_{negative}\}$ of current candidate news are followed, which augment the LLM's understanding of this task. Finally, we prompt the LLM to predict the result of current news and give its corresponding supportive explanation.

In summary, the inside judge process can be expressed as:

$$P_i, R_i = \mathcal{F}(Ins_i \uplus \mathcal{II} \uplus x_{test}), \quad (4)$$

where Ins_i is the inside instruction, $\uplus$ denotes the concatenation operation of two textual pieces, and x_{test} represents the current candidate news. P_i refers to the prediction result, while R_i denotes the corresponding explanation. You can move to Appendix A.2 for more details about this prompt.

Outside Judge. With the outside investigation information $\mathcal{OI}$ retrieved through the google search

[2]Here, the training set refers to the few-shot training data.

[3]This approach ensures the diversity and effectiveness of the retrieved samples.

[4]https://serpapi.com/

engine as described in Section 4.2, we can derive the outside judge prediction, which is crucial for real-time news detection.

Similar to the design of Inside Judge, we describe the objective of fake news detection through an outside instruction, followed by the candidate news to be detected and the retrieved outside investigation documents $\mathcal{OI}$. After that, we can derive the outside prediction P_o and explanation R_o:

$$P_o, R_o = \mathcal{F}(Ins_o \uplus x_{test} \uplus \mathcal{OI}), \quad (5)$$

where Ins_o denotes the outside instruction. Detailed information about this prompt is available in Appendix A.3.

Following these two parallel judge processes, we obtain the prediction results P_i, P_o and corresponding explanations R_i, R_o from inside and outside perspectives, respectively.

4.4 Determination Module

With the predictions P_i, P_o and their corresponding explanations R_i, R_o, the final outputs are obtained by jointly considering these two perspectives.

More specifically, if the two predictions are identical (i.e., $P_i = P_o$), we can directly derive the final prediction with high confidence. Nevertheless, if two results diverge, indicating a conflict between the Inside Judge and Outside Judge, we further propose a determination selector to make a choice based on both sets of predictions and explanations:

$$P_f = \mathcal{F}(Ins_d \uplus x_{test} \uplus P_i \uplus R_i \uplus P_o \uplus R_o), \quad (6)$$

where Ins_d denotes the determination instruction (see Appendix A.4 for more details). P_f is the final inference result of the DAFND model.

5 Experiments

5.1 Experiment Setup

Datasets and Evaluation Metrics. We conduct experiments on two datasets, *PolitiFact* and *Gossipcop*, both of which are proposed in a benchmark called FakeNewsNet (Shu et al., 2020). PolitiFact consists of various political news, while Gossipcop is sourced from an entertainment story fact-checking website. For the few-shot setting, following the strategy employed in (Jiang et al., 2022; Ma et al., 2023), we randomly select $K \in (8, 32, 100)$ positive and negative news articles as the training set, respectively. More statistics about the datasets are illustrated in Table 1.

Dataset		PolitiFact	Gossipcop
Train	# True news	8/32/100	8/32/100
	# Fake news	8/32/100	8/32/100
	# Total news	16/64/200	16/64/200
Test	# True news	120	3,200
	# Fake news	80	1,060
	# Total news	200	4,260

Table 1: Statistics of PolitiFact and Gossipcop datasets.

Given that the task focuses on detecting fake news, fake news articles are regarded as positive examples (Ma et al., 2023). We further adopt the F1-score and Accuracy (ACC) as the evaluation metrics to measure classification performance.

Implementation Details. In DAFND architecture, we utilize the *zephyr-7b-beta* (Tunstall et al., 2023) model on Huggingface as the LLM to conduct experiments. When running Zephyr, we adhere to the default parameter values provided by the official, where the sampling temperature is 0.70, top_k is 50, and top_p is 0.95. The max_new_token is set to 256, and do_sample is set as $True$.

In the Detection Module (Section 4.1), the number of keywords to extract is set to $N = 5$. In the Inside Investigation part (Section 4.2), we employ the *DeBERTa-base* model (He et al., 2021) from Transformers (Wolf et al., 2020) as the representation model. The number of retrieved positive/negative nearest neighbors is set as $k = 2$.

All experiments are conducted on a Linux server with two 3.00GHz Intel Xeon Gold 5317 CPUs and two Tesla A100 GPUs.

Benchmark Methods. In order to verify the effectiveness of our DAFND model, we compare DAFND with the state-of-the-art few-shot fake news detection methods. According to the model architecture, they can be grouped into three categories, including traditional fake news detection methods (① ∼ ④), LLM-based methods (⑤ ∼ ⑧), and hybrid methods (⑨).

- ① **PROPANEWS** (Huang et al., 2023) proposes a novel framework for generating more valuable training examples, which is more beneficial to human-written situations[5].

[5]Specifically, We combine the few-shot training set and the augmented data to train a RoBERTA-Large-based classifier,

Dataset	Methods	ACC			F-1 score		
		K=8	K=32	K=100	K=8	K=32	K=100
PolitiFact	① PROPANEWS	40.00	43.50	40.00	57.14	58.30	57.14
	② FakeFlow	61.00	62.50	63.50	44.29	47.55	48.95
	③ MDFEND	65.50	64.00	71.50	62.30	64.36	69.84
	④ PSM	70.00	72.50	79.00	49.15	52.38	65.70
	⑤ Zephyr	60.00	63.50	66.50	48.72	53.50	54.42
	⑥ ChatGLM-3	68.50	68.50	72.50	58.28	58.82	64.05
	⑦ LLama-3	69.50	70.50	69.00	63.91	65.09	64.00
	⑧ GPT-3.5	71.00	69.50	73.00	60.27	60.65	64.47
	⑨ ARG	74.00	78.50	82.50	67.16	68.61	80.61
	DAFND (ours)	**86.50**	**87.50**	**88.50**	**81.40**	**82.80**	**84.40**
Gossipcop	① PROPANEWS	24.88	25.40	24.88	39.85	39.97	39.85
	② FakeFlow	57.89	58.26	57.28	26.60	27.66	28.18
	③ MDFEND	41.27	56.08	63.73	40.20	42.06	44.52
	④ PSM	77.44	78.05	78.30	41.73	41.37	54.20
	⑤ Zephyr	67.21	65.85	67.23	27.05	27.43	27.67
	⑥ ChatGLM-3	62.49	62.75	63.43	31.59	34.83	34.15
	⑦ LLama-3	65.96	65.85	66.17	30.89	35.07	31.74
	⑧ GPT-3.5	68.50	69.44	67.44	32.90	36.73	36.58
	⑨ ARG	61.41	77.42	76.50	42.32	51.46	46.57
	DAFND (ours)	**82.80**	**82.80**	**83.30**	**53.50**	**55.00**	**56.50**

Table 2: Experimental results of our proposed method on the PolitiFact and Gossipcop datasets. **Bold font** represents the optimal result. For baseline methods, we follow their publicly released codes to obtain the results.

- ② **FakeFlow** (Ghanem et al., 2021) devises a model that detects fake news articles by integrating the flow of affective information.
- ③ **MDFEND** (Nan et al., 2021) incorporates the domain information through a domain gate mechanism to aggregate multiple representations extracted by a mixture of experts.
- ④ **PSM** (Ni et al., 2020) proposes to utilize Propensity Score Matching (PSM) to select deconfounded features, thereby boosting the detection of fake news.
- ⑤ **Zephyr** (Tunstall et al., 2023) represents the advanced 7B model, which is optimized by the preference data from AI Feedback.
- ⑥ **ChatGLM-3** (Zeng et al., 2022; Du et al., 2022) is a series of pre-trained dialogue models, and we select the ChatGLM3-6B version for the experimental comparison.
- ⑦ **LLama-3** (AI@Meta, 2024) refers to the LLM proposed by Meta. We adopt its 8B version (Meta-Llama-3-8B-Instruct) sourced from Huggingface for experiments.
- ⑧ **GPT-3.5** (Ouyang et al., 2022) is an advanced LLM developed by OpenAI. We leverage the API (version: gpt-3.5-turbo-0613) to conduct in-context learning.
- ⑨ **ARG** (Hu et al., 2024) designs an adaptive rationale guidance network for fake news detection, which integrates insights from both LLMs and traditional detection methods.

It is worth noting that, for these LLM-based baselines (⑤ $\sim$ ⑧), we adhere to the instruction prompt proposed by (Hu et al., 2024) to conduct in-context learning. Besides, due to the limitations of maximum tokens, we randomly select 6 samples as the demonstrations, which is more than the demonstration samples utilized in our DAFND model[6], facilitating a fair comparison[7].

adhering to the setting in the original paper.

[6]As introduced in Section 5.1, DAFND utilizes $k = 2$ positive and negative samples as demonstrations, with a total number of 4.

[7]For LLM-based baselines, if the LLM fails to make an inference or we are unable to categorize the output as *Real* or *Fake* categories, we directly treat it as the wrong prediction.

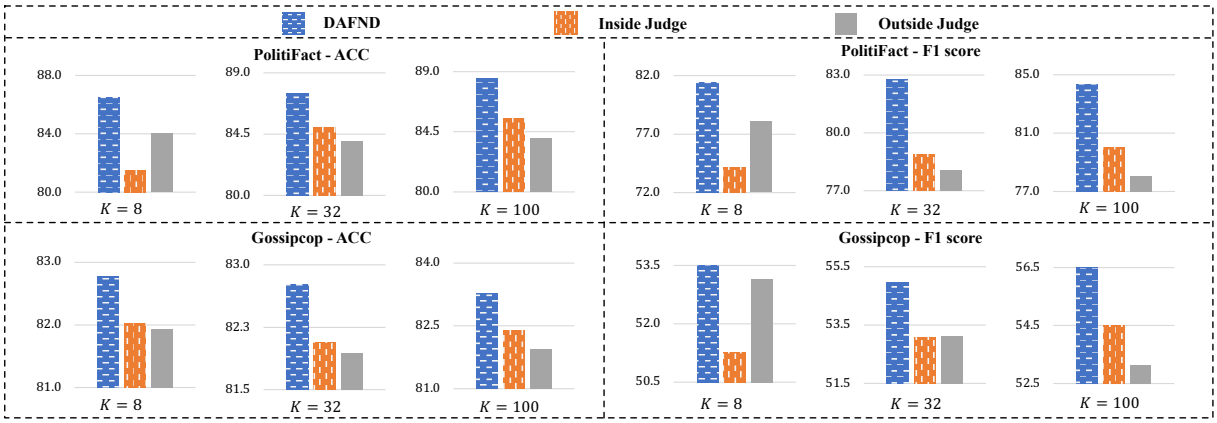


Figure 3: Ablation experiments on the PolitiFact and Gossipcop dataset.

5.2 Experimental Result

The main results, presented in Table 2, indicate that our proposed DAFND model surpasses all baselines across various metrics, encompassing traditional, LLM-based and hybrid methods. This underscores the effectiveness of our design and the advantages of enhancing the LLM through both inside and outside perspectives. Furthermore, several notable phenomena emerge from these results:

Firstly, for most baselines and our DAFND model, the performance on the PolitiFact dataset exceeds that on the Gossipcop dataset, suggesting that Gossipcop presents greater difficulty. Specifically, PolitiFact consists of political news while Gossipcop pertains to the entertainment domain. This disparity is reasonable as political news typically exhibits more organized format and content, which facilitates the fake news detection process. Secondly, with the increase of training instances (K), most traditional fake news detection methods (e.g., ④ PSM) and hybrid methods (⑨ ARG) show improved performance. This is logical as more data enables better training of a supervised model, mitigating the lack of prior knowledge. However, an exception is observed in ① PROPANEWS, whose performance appears relatively unaffected by K. As introduced in Section 5.1, different from other methods, PROPANEWS designs a data constructing strategy to supplement original training set. This significantly offsets the impact of training data quantity. Moreover, for LLM-based methods (⑤ $\sim$ ⑧), as outlined in Section 5.1, due to the limitation of maximum tokens, we all randomly select 6 samples as demonstrations. Hence, increasing the number of training instances does not substantially benefit the in-context learning of LLMs. Thirdly, although our DAFND model also faces the constraint of maximum token limitations, the kNN retrieval mechanism in the Inside Investigation part (Section 4.2) enhances the utilization of increased training data, thereby achieving a certain degree of improvements with higher K values. Fourthly, the hybrid method (⑨ ARG), benefiting from the joint modeling of traditional models and LLMs, obtains competitive performance. And compared to that, DAFND still maintains a significant advantage, particularly in scenarios with scarcer data. These observations further demonstrate the effectiveness of our designs from multiple perspectives.

5.3 Ablation Study

In this subsection, we conduct ablation experiments to assess the effectiveness of different components within our model. Specifically, we directly remove the Determination Module, consequently leading to two ablated variants: Inside Judge and Outside Judge. The results are depicted in Figure 3.

From this figure, across all configurations on PolitiFact and Gossipcop datasets, noticeable decreases are observed between the full DAFND model and its two ablated variants. This thoroughly substantiates the essentiality and non-redundancy of our DAFND designs. Notably, when the training instances are relatively limited ($K = 8$), Inside Judge exhibits less effectiveness compared to Outside Judge. With the increase of training instances ($K = 32, 100$), Inside Judge progressively achieves more competitive performance than Outside Judge. This can be attributed to the increasing benefit of additional training data for the Inside Investigation and Inside Judge components. In contrast, Outside Investigation and Outside Judge rely on the retrieved information online, which remains

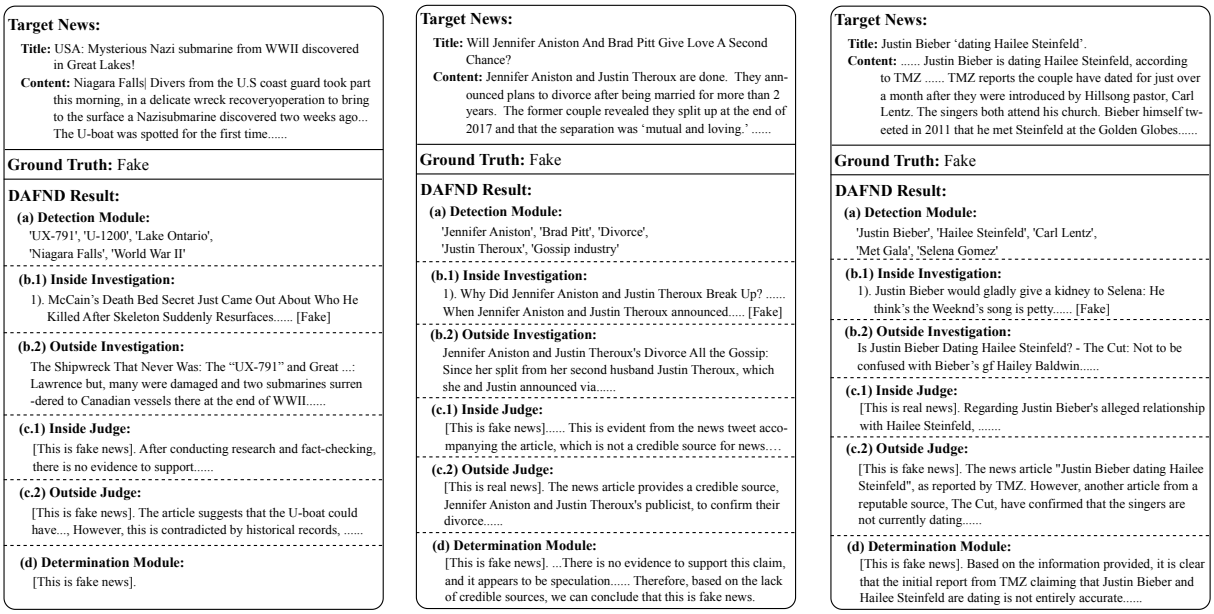


Figure 4: The case study of the DAFND model. Specifically, (a) is from the PolitiFact dataset (K=100), while (b) and (c) are from the Gossipcop dataset (K=100).

unaffected by the quantity of training data.

Furthermore, considering that the backbone of DAFND is ⑤ Zephyr (Tunstall et al., 2023), we can jointly compare the results in Figure 3 and the "⑤ Zephyr" line in Table 2. This comparison highlights that both Inside Judge and Outside Judge significantly outperform Zephyr, thereby demonstrating the efficacy of our motivation to design DAFND from both inside and outside perspectives.

5.4 Case Study

To further illustrate the effectiveness of different modules in our model, we conduct a case study on both PolitiFact and Gossipcop datasets. Specifically, Figure 4 presents the input information (i.e., target news), the ground truth label, DAFND results (including the intermediate results of each module and the final results).

As depicted Figure 4 (a), the Detection Module accurately identifies the key information within the target news. And based on the valid results from Inside Investigation and Outside Investigation, both the Inside Judge and Outside Judge correctly infer that: "*[This is fake news].*". These are fed into the Determination Module, leading to the final prediction, which is consistent with the ground truth label (i.e., *[Fake]*). Moreover, In Figure 4 (b), Inside Judge obtains the right inference (i.e., *[Fake]*), while Outside Judge makes the wrong inference (i.e., *[Real]*). Conversely, in Figure 4 (c), Inside Judge incorrectly infers "*[Real]*", while Outside Judge accurately predicts "*[Fake]*". In both (b) and (c), with the design of Determination Module, DAFND finally makes the correct decision. These cases intuitively demonstrate the significant role of each module in DAFND, affirming its efficacy.

More experimental analyses, such as Bad Case Analysis, can be found in Appendix B.

6 Conclusions

In this paper, we explored a motivated direction for few-shot fake news detection. We began by analyzing the limitations of current LLM-based detection methods, identifying two primary challenges: (1) Understanding Ambiguity and (2) Information Scarcity. To address these issues, we developed a Dual-perspective Augmented Fake News Detection (DAFND) model. In DAFND, a Detection Module was designed to identify keywords from the given news. Then, we proposed an Investigation Module, a Judge Module to retrieve valuable information and further generate respective predictions. More importantly, a Determination Module integrated the predictions from both inside and outside perspectives to produce the final output. Finally, extensive experiments on two publicly available datasets demonstrated the effectiveness of our proposed method. We hope our work will lead to more future studies.

Limitations

In our proposed DAFND method, we need to integrate the LLM (i.e., *zephyr-7b-beta* introduced in Section 5.1). Due to the large scale of LLMs, it tends to consume more computing resources and time compared to traditional baselines, such as PSM (Ni et al., 2020) and FakeFlow (Ghanem et al., 2021). Essentially speaking, LLMs contain a vast amount of knowledge, much of which may be unnecessary for fake news detection. Distilling useful knowledge so as to accelerate the inference remains a valuable and intriguing research direction.

Another limitation is that our current approach only employs LLMs for inference. Although we design precise prompts to implement in-context learning, it still cannot fully exploit the capabilities of LLMs due to the inherent gap between the natural language and the knowledge encoded in the model parameters. In future work, we would like to explore the low resource scenario fine-tuning techniques (e.g., lora (Hu et al., 2021a)) to adapt LLMs for the few-shot fake news detection task.

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A Prompts

In this section, we illustrate the prompts utilized in our DAFND methodology, which can serve as a valuable resource for future research in this area.

A.1 Detection Prompt

As a news keyword extractor, your task is to extract the N most important keywords from a given news text. The keywords should include when, where, who, what, how and why the news happened. Please give me the six keywords only. My first suggestion request is ***{Target News Document}***.

A.2 Inside Judge Prompt

I need your assistance in evaluating the authenticity of a news article. I will provide you the news article and additional information about this news. You have to answer that [This is fake news] or [This is real news] in the first sentence of your output and give your explanation about [target news].

I will give you some examples of news. Your answer after [output] should be consistent with the following examples:

[example 1]:

[input news]: [news title: ***{...}****, news text:* ***{...}****, news tweet:* ***{...}****]*

[output]: [This is ***{...}*** *news]*

.....

[target news]:

[input news]: [news title: ***{...}****, news text:* ***{...}****, news tweet:* ***{...}****]*

[output]:

A.3 Outside Judge Prompt

I need your assistance in evaluating the authenticity of a news article. I will provide you the news article and additional information about this news. Please analyze the following news and give your decision. The first sentence of your [Decision] must be [This is fake news] or [This is real news].

The news article is: ***{...}****.*

The additional information is: ***{...}****.*

[Decision]:

A.4 Determination Prompt

I need your assistance in evaluating the authenticity of a news article. This news article include news title, news text and news tweet.

The news article is: news title: ***{...}****, news text:* ***{...}****, news tweet:* ***{...}****.*

There are two different views on this news article.

Some people believe that ***{...}****, their explanation is:* ***{...}****.*

Others believe that ***{...}****, their explanation is:* ***{...}****.*

Please judge their opinion and give your decision. The first sentence after [Explanation] must be [This is fake news] or [This is real news], and then give your explanation.

[Explanation]:

B Bad Case Analysis

In this section, we illustrate the bad cases that DAFND struggles with, with a goal to analyze its shortcomings and possible improvement directions.

Target News:

Title: Stephen Colbert Promises an Open Bar for the Losers at the Emmys!

Content: ..... There's never been booze. The first-time host has vowed to change that—or at least he wishes he could. With the Emmys fast approaching, Colbert recently discussed his plans for Sunday\'s ceremony with E! News. The comedian said ... he is considering having a rolling bar for the losers at.....

Ground Truth: Real

DAFND Result:

(a) Detection Module:
'Stephen Colbert', 'Emmys', 'Rolling bar', 'Losers', 'No bar'

(b.1) Inside Investigation:
1). Nina Dobrev and Orlando Bloom Are "Casually Dating"..... The 28-year-old Vampire Diaries star was..... [Fake]

(b.2) Outside Investigation:
2017 Emmys Host Stephen Colbert Promises an Open Bar..... : Listen up, Emmys nominees! Stephen Colbert is offering up a pretty sweet consolation prize to the evening's losers.

(c.1) Inside Judge:
[This is fake news]..... There is no evidence to support the claim that this will actually happen.....

(c.2) Outside Judge:
[This is fake news]. Although the news article and the additional information all come from reputable sources, ... Until there is any further confirmation, it's best to treat this news as unverified and potentially false.

(d) Determination Module:
[This is fake news].

Figure 5: The bad case of DAFND on the Gossipcop dataset (K=100).

As illustrated in Figure 5, a particular failure occurs when there are no relevant samples in the few-shot training set for retrieval. Consequently, the Inside Investigation and Inside Judge modules fail to work effectively. Meanwhile, although Out-

side Investigation retrieves a proof supporting the given news, Outside Judge still classifies it into *Fake* category because it claims the necessity of further confirmation. This bad case reveals two potential directions for improving our model:

For one thing, as introduced in Section 1, DAFND retrieves similar demonstrations from the training set in response to the Understanding Ambiguity problem. This relies on the assumption that valuable samples can be found in the training set, which is not always the case. As a consequence, the failure in Figure 5 occurs. Therefore, how to discriminate and mitigate the impact of such circumstances is a crucial direction for improving the DAFND design.

For another, as discussed in the Limitation section, we only employ the LLM to conduct inference, which cannot fully exploit LLMs' powerful capabilities. In Figure 5, although Outside Investigation retrieves valuable information, the prediction of Outside Judge still goes wrong. In fact, despite the great reasoning ability of these general LLMs, they are not competent in news-related domains and are not sufficiently familiar with the specific expression characteristics. Thence, we would like to adopt fine-tuning techniques to adapt LLMs for the news corpus, which we believe could bring a positive effect to DAFND.